

Long Division

Every graded set, every week — 4 weeks, 20 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

How Many Fit

Estimate before you divide anything.

✦ Warm-Up 6 ITEMS

How many 10s in 80

How many 20s in 60

How many 30s in 90

$90 \div 30$

$120 \div 40$

$200 \div 50$

◆ Core GRADED • SHOW YOUR WORKING

1. About how many 30s in 400 (13 thirties is 390)

2. $600 \div 20$

3. $840 \div 40$

4. $960 \div 30$

5. $1200 \div 60$

★ Challenge NOT GRADED

C1. About how many 48s in 2,400

C2. $7,200 \div 90$

Long Division

Divide, multiply, subtract, bring down. Every answer here comes out whole.

 Warm-Up 6 ITEMS

84 ÷ 12

96 ÷ 16

75 ÷ 15

91 ÷ 13

144 ÷ 12

156 ÷ 13

 Core GRADED • SHOW YOUR WORKING

1. 552 ÷ 24

.....

2. 784 ÷ 28

.....

3. 945 ÷ 35

.....

4. 1,224 ÷ 36

.....

5. 1,472 ÷ 32

.....

 Challenge NOT GRADED

C1. 3,036 ÷ 23

.....

C2. 5,152 ÷ 46

.....

Four Digits

Longer, not harder. Watch the place where the quotient has a zero.

✦ Warm-Up 6 ITEMS

$1,000 \div 10$

$2,400 \div 12$

$3,600 \div 12$

$4,200 \div 20$

$5,000 \div 25$

$6,300 \div 30$

◆ Core GRADED • SHOW YOUR WORKING

1. $3,384 \div 24$

2. $4,536 \div 21$

3. $7,308 \div 36$ (Note the zero in the middle)

4. $8,464 \div 46$

5. $9,072 \div 24$

★ Challenge NOT GRADED

C1. $9,594 \div 26$

.....

C2. $6,120 \div 24$

.....

Remainders

The situation decides what happens to the leftover.

Warm-Up 6 ITEMS

17 ÷ 5 — the remainder

17 ÷ 5 — the whole-number quotient

23 ÷ 4 — remainder

50 ÷ 7 — remainder

100 ÷ 9 — remainder

45 ÷ 6 — remainder

Core GRADED • SHOW YOUR WORKING

1. 127 children, 30 per bus. Buses needed (You cannot leave seven behind)
2. 127 ÷ 30 — the remainder
3. \$100 shared by 8 — dollars each, rounded down
4. 95 eggs into boxes of 12 — full boxes
5. 95 eggs into boxes of 12 — eggs left over

Challenge NOT GRADED

C1. 17 ÷ 4 as a mixed number — type the fraction part as a/b

C2. A rope of 250 cm cut into 40 cm pieces. Whole pieces

Quotient Hunt & Road Trip

Enrichment. Estimate out loud before computing.

✦ Warm-Up 2 ITEMS

600 miles over 3 days — miles a day

\$120 for 4 days of food — dollars a day

◆ Core GRADED • SHOW YOUR WORKING

1. 1,200 miles at 60 mph — hours driving

2. A tank does 400 miles. Tanks needed for 1,000 miles (Two and a half is not enough)

★ Challenge NOT GRADED

C1. \$1,440 shared over 12 days

C2. 960 miles over 4 days, 8 hours a day — average mph

C3. Estimate $4,850 \div 48$ to the nearest ten

Week 1 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • How Many Fit

OUT OF 11

Warm-Up • 8 | 3 | 3 | 3 | 3 | 4

Core • 13 | 30 | 21 | 32 | 20

Challenge • 50 | 80

1.2 • Long Division

OUT OF 11

Warm-Up • 7 | 6 | 5 | 7 | 12 | 12

Core • 23 | 28 | 27 | 34 | 46

Challenge • 132 | 112

1.3 • Four Digits

OUT OF 11

Warm-Up • 100 | 200 | 300 | 210 | 200 | 210

Core • 141 | 216 | 203 | 184 | 378

Challenge • 369 | 255

1.4 • Remainders

OUT OF 11

Warm-Up • 2 | 3 | 3 | 1 | 1 | 3

Core • 5 | 7 | 12 | 7 | 11

Challenge • 1/4 | 6

Fri • Quotient Hunt & Road Trip

OUT OF 4

Warm-Up • 200 | 30

Core • 20 | 3

Challenge • 120 | 30 | 100

Four Digits by Two

Where the first digit of the quotient goes, and why.

✦ Warm-Up 6 ITEMS

$1200 \div 12$

$2400 \div 12$

$3600 \div 12$

$1000 \div 10$

$4200 \div 20$

$5000 \div 25$

◆ Core GRADED • SHOW YOUR WORKING

1. $4536 \div 21$

2. $8464 \div 46$

3. $9072 \div 24$

4. $3384 \div 24$

5. $6120 \div 24$

★ Challenge NOT GRADED

C1. $9594 \div 26$

.....

C2. $5152 \div 46$

.....

Zeros in the Quotient

The place that gets skipped is the place people lose.

✦ Warm-Up 6 ITEMS

$600 \div 6$

$800 \div 8$

$606 \div 6$

$404 \div 4$

$900 \div 9$

$1000 \div 5$

◆ Core GRADED • SHOW YOUR WORKING

1. $7308 \div 36$ _____

2. $4080 \div 8$ _____

3. $6036 \div 6$ _____

4. $2050 \div 25$ _____

5. $5025 \div 25$ _____

★ Challenge NOT GRADED

C1. $12060 \div 60$

C2. $8064 \div 32$

Estimate the Quotient

A rough answer means a wrong one gets noticed.

✦ Warm-Up 6 ITEMS

Estimate $400 \div 20$

Estimate $600 \div 30$

Estimate $900 \div 30$

Estimate $1200 \div 40$

Estimate $800 \div 20$

Estimate $1000 \div 50$

◆ Core GRADED • SHOW YOUR WORKING

1. Estimate $4850 \div 48$ to the nearest ten

.....

2. Estimate $3120 \div 39$

.....

3. Estimate $6300 \div 68$ to the nearest ten

.....

4. Estimate $2450 \div 51$

.....

5. Estimate $7800 \div 19$ to the nearest hundred

.....

★ Challenge NOT GRADED

C1. True value of $3120 \div 39$

.....

C2. True value of $4850 \div 50$

.....

Divisors Near a Ten

Rounding 29 to 30 makes the guessing quick.

✦ Warm-Up 6 ITEMS

$600 \div 30$

$620 \div 20$

$900 \div 30$

$840 \div 40$

$960 \div 30$

$720 \div 20$

◆ Core GRADED • SHOW YOUR WORKING

1. $812 \div 29$

2. $1178 \div 31$

3. $1449 \div 21$

4. $2451 \div 43$

5. $1764 \div 42$

★ Challenge NOT GRADED

C1. $3196 \div 47$

.....

C2. $5688 \div 79$

.....

Road Trip Budget Begins

Pick the route, gather the real numbers.

 Warm-Up 2 ITEMS

600 miles over 3 days — miles a day

\$120 over 4 days — dollars a day

 Core GRADED • SHOW YOUR WORKING

1. 1200 miles at 60 mph — hours
.....

2. A tank does 400 miles — tanks for 1000 miles
.....

 Challenge NOT GRADED

C1. \$1440 over 12 days
.....

C2. 960 miles over 4 days, 8 hours a day — average mph
.....

C3. 2400 miles at 55 mph — hours, rounded up
.....

Week 2 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Four Digits by Two

OUT OF 11

Warm-Up • 100 | 200 | 300 | 100 | 210 | 200

Core • 216 | 184 | 378 | 141 | 255

Challenge • 369 | 112

2.2 • Zeros in the Quotient

OUT OF 11

Warm-Up • 100 | 100 | 101 | 101 | 100 | 200

Core • 203 | 510 | 1006 | 82 | 201

Challenge • 201 | 252

2.3 • Estimate the Quotient

OUT OF 11

Warm-Up • 20 | 20 | 30 | 30 | 40 | 20

Core • 100 | 80 | 90 | 50 | 400

Challenge • 80 | 97

2.4 • Divisors Near a Ten

OUT OF 11

Warm-Up • 20 | 31 | 30 | 21 | 32 | 36

Core • 28 | 38 | 69 | 57 | 42

Challenge • 68 | 72

Fri • Road Trip Budget Begins

OUT OF 4

Warm-Up • 200 | 30

Core • 20 | 3

Challenge • 120 | 30 | 44

Find the Remainder

It is always smaller than the divisor. Always.

Warm-Up 6 ITEMS

$17 \div 5$ — the remainder

$23 \div 4$ — the remainder

$50 \div 6$ — the remainder

$100 \div 3$ — the remainder

$20 \div 5$ — the remainder

$19 \div 6$ — the remainder

Core GRADED • SHOW YOUR WORKING

1. $100 \div 7$ — the remainder

.....

2. $500 \div 23$ — the remainder

.....

3. $750 \div 24$ — the remainder

.....

4. $1000 \div 31$ — the remainder

.....

5. $250 \div 12$ — the remainder

.....

Challenge NOT GRADED

C1. Dividing by 23, the remainder must be less than

.....

C2. The largest possible remainder when dividing by 24

.....

Round Up or Down

Buses round up. Full boxes round down.

+ Warm-Up 6 ITEMS

100 people, buses hold 30 — buses

100 pencils, boxes of 30 — full boxes

50 people, cars hold 4 — cars

50 pencils, packs of 4 — full packs

10 people, tables of 4 — tables

10 pencils, packs of 4 — full packs

◆ Core GRADED · SHOW YOUR WORKING

1. 200 people, buses hold 45 — buses

.....

2. 200 items, boxes of 45 — full boxes

.....

3. 1,000 miles at 200 a day — days

.....

4. 1,050 miles at 200 a day — days

.....

5. 90 cupcakes, trays of 12 — trays

.....

★ Challenge NOT GRADED

C1. 1,200 miles with no day over 200 — days

.....

C2. 145 people, buses hold 30 — buses

.....

Keep It as a Fraction

Pizza does not round. Neither does money.

✦ Warm-Up 6 ITEMS

\$10 split 4 ways

\$5 split 2 ways

\$9 split 2 ways

\$20 split 8 ways

\$6 split 4 ways

\$12 split 8 ways

◆ Core GRADED • SHOW YOUR WORKING

1. \$50 split 4 ways, to the cent

2. \$100 split 30 ways, to the cent

3. \$40 split 3 ways, to the cent

4. 7 pizzas split 4 ways, as a decimal

5. \$15 split 4 ways

★ Challenge NOT GRADED

C1. $100 \div 7$ as a decimal to two places

C2. \$100 split 30 ways — cents left over

Four Questions, One Division

$100 \div 30$ answered four ways, all correct.

Warm-Up 6 ITEMS

$100 \div 30$ — the whole part

$100 \div 30$ — the remainder

$100 \div 30$ rounded up

$100 \div 30$ rounded down

30×3

$100 - 90$

Core GRADED • SHOW YOUR WORKING

1. 100 people, buses of 30 — buses

2. 100 pencils, full boxes of 30

3. \$100 split 30 ways, to the cent

4. $100 \div 30$ — the remainder

5. $30 \times 3 + 10$

Challenge NOT GRADED

C1. For money, which answer suits — type decimal or remainder

C2. For buses, which suits — type up or down

Budget the Miles

Miles per day, tanks per trip, dollars per person.

 Warm-Up 6 ITEMS

600 miles over 3 days — miles a day

\$120 over 4 days — dollars a day

1200 ÷ 6

400 × 3

60 × 20

240 × 5

 Core GRADED • SHOW YOUR WORKING

1. 1,200 miles at 60 mph — hours

2. A tank does 400 miles — tanks for 1,200

3. \$1,440 over 12 days

4. 960 miles over 4 days

5. 2,400 miles at 55 mph — hours, rounded up

 Challenge NOT GRADED

C1. 1,200 miles over 5 equal days — miles a day

C2. Capped at 200 miles a day — days

Week 3 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Find the Remainder

OUT OF 11

Warm-Up • 2 | 3 | 2 | 1 | 0 | 1

Core • 2 | 17 | 6 | 8 | 10

Challenge • 23 | 23

3.2 • Round Up or Down

OUT OF 11

Warm-Up • 4 | 3 | 13 | 12 | 3 | 2

Core • 5 | 4 | 5 | 6 | 8

Challenge • 6 | 5

3.3 • Keep It as a Fraction

OUT OF 11

Warm-Up • 2.5 | 2.5 | 4.5 | 2.5 | 1.5 | 1.5

Core • 12.5 | 3.33 | 13.33 | 1.75 | 3.75

Challenge • 14.29 | 10

3.4 • Four Questions, One Division

OUT OF 11

Warm-Up • 3 | 10 | 4 | 3 | 90 | 10

Core • 4 | 3 | 3.33 | 10 | 100

Challenge • decimal | up

Fri • Budget the Miles

OUT OF 11

Warm-Up • 200 | 30 | 200 | 1200 | 1200 | 1200

Core • 20 | 3 | 120 | 240 | 44

Challenge • 240 | 6

Finish the Budget

Every line item divided out and totalled.

 Warm-Up 6 ITEMS

3 days at \$120

\$360 + \$200

5 × \$40

\$600 – \$560

4 × \$50

\$1,000 ÷ 5

 Core GRADED • SHOW YOUR WORKING

1. 12 days at \$120
2. 3 tanks at \$65
3. \$1,440 + \$195
4. \$1,635 split 4 people, to the cent
5. 1,200 miles over 3 tanks

 Challenge NOT GRADED

C1. \$1,635 over 12 days, to the cent

C2. A \$2,000 budget less \$1,635

Defend the Rounding

Pick three lines and say why each rounded the way it did.

✦ Warm-Up 6 ITEMS

2.9 tanks of fuel — tanks bought

2.9 rounded down

Round 13.334 to the cent

Round 2.5 up

Round 2.4 down

Round \$9.996 to the cent

◆ Core GRADED • SHOW YOUR WORKING

1. 1,250 miles, a tank does 400 — tanks needed

.....

2. 1,250 miles — full tanks used

.....

3. \$50 split 3 ways, to the cent

.....

4. Cents left over on that split

.....

5. $1,200 \div 400$

.....

★ Challenge NOT GRADED

C1. 1,250 miles with 4 tanks bought — miles of spare range

.....

C2. \$1,635 split 12 ways, to the cent

.....

Mixed Review

Long division and remainder interpretation together.

 Warm-Up 6 ITEMS

$84 \div 12$

$372 \div 3$

$100 \div 7$ — the whole part

$100 \div 7$ — the remainder

$600 \div 30$

$936 \div 24$

 Core GRADED • SHOW YOUR WORKING

1. $4,536 \div 21$

2. $7,308 \div 36$

3. $812 \div 29$

4. $8,464 \div 46$

5. $750 \div 24$ — the remainder

 Challenge NOT GRADED

C1. $12,060 \div 60$

.....

C2. 100 people, buses of 30 — buses

.....

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

✦ Warm-Up 6 ITEMS

$72 \div 8$

$618 \div 6$

$63 \div 9$

$300 \div 5$

$120 \div 4$

$360 \div 6$

◆ Core GRADED • SHOW YOUR WORKING

1. $936 \div 24$

2. $3,120 \div 39$

3. $1,178 \div 31$

4. $9,072 \div 24$

5. $2,451 \div 43$

★ Challenge NOT GRADED

C1. $6,036 \div 6$

.....

C2. $500 \div 23$ — the remainder

.....

Mission 03 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 2 ITEMS

$618 \div 6$

$884 \div 34$

Core GRADED • SHOW YOUR WORKING

1. $8,464 \div 46$

2. $9,072 \div 24$

3. $6,036 \div 6$

4. $1,178 \div 31$

5. $750 \div 24$ — the remainder

6. Estimate $6,300 \div 68$ to the nearest ten

7. 100 people, buses of 30 — buses

8. 100 pencils, full boxes of 30

Challenge NOT GRADED

C1. \$100 split 30 ways, to the cent
.....

C2. 1,200 miles capped at 200 a day — days
.....

Week 4 Answers

Mission 03 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Finish the Budget

OUT OF 11

Warm-Up • 360 | 560 | 200 | 40 | 200 | 200

Core • 1440 | 195 | 1635 | 408.75 | 400

Challenge • 136.25 | 365

4.2 • Defend the Rounding

OUT OF 11

Warm-Up • 3 | 2 | 13.33 | 3 | 2 | 10

Core • 4 | 3 | 16.67 | 1 | 3

Challenge • 350 | 136.25

4.3 • Mixed Review

OUT OF 11

Warm-Up • 7 | 124 | 14 | 2 | 20 | 39

Core • 216 | 203 | 28 | 184 | 6

Challenge • 201 | 4

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • 9 | 103 | 7 | 60 | 30 | 60

Core • 39 | 80 | 38 | 378 | 57

Challenge • 1006 | 17

Fri • Mission 03 Test

OUT OF 10

Warm-Up • 103 | 26

Core • 184 | 378 | 1006 | 38 | 6 | 90 | 4 | 3

Challenge • 3.33 | 6